

# Component-Specific Troubleshooting Theory

## Task 1: RAM Reseating Simulation & Documentation

- **Step-by-Step Procedure:**

1. **Safety & Power Down:** Shut down the computer completely, unplug the AC power cable from the wall, and press and hold the power button for 15 seconds to drain residual charge from motherboard capacitors.
2. **ESD Protection & Opening Case:** Wear an anti-static wrist strap (or touch an unpainted metal section of the case) and remove the side panel to access the motherboard.
3. **Locate RAM Slots & Unlatch:** Locate the DIMM slots next to the CPU socket. Press down on the locking latches at both ends of the RAM slot (or the single active latch on modern boards) until the RAM stick pops up slightly.
4. **Remove & Inspect:** Lift the module straight out by its edges without touching the golden contact pins. Inspect the contacts and slot for dust or debris (clean contacts with an eraser or isopropyl alcohol if dirty).
5. **Re-align & Re-insert:** Align the notch on the bottom of the RAM module with the plastic key in the DIMM slot (ensuring it is oriented in the correct direction).
6. **Apply Even Pressure:** Place both thumbs on the top outer corners of the RAM stick and press firmly straight down until the side latches snap into place with a clear, audible "click."

- **Observations & Potential Issues Encountered:**

- *Key Alignment:* The notch is off-center; attempting to insert it backward will damage the slot.
- *Partial Seating:* If uneven pressure is applied, one side may click while the other remains raised, resulting in a black screen and three POST beeps or debug code 55/C1.

## Task 2: Troubleshooting Checklist for a Faulty GPU (5 Steps)

- **Step 1: Symptom Identification**
  - Observe the visual defect: Is there complete "No Signal" on the monitor, screen flickering, artifacting (colored lines/checkered blocks), or display driver crashing during 3D gaming?
- **Step 2: External Connection & Cable Verification**
  - Ensure the display cable (HDMI or DisplayPort) is plugged directly into the **GPU ports**, not the motherboard video output. Test with a known-good cable and alternate monitor.
- **Step 3: Physical Inspection & Hardware Reseating**
  - Power off the system, unmount the GPU, inspect the PCIe slot, clean gold contacts, and reseat the card firmly into the top primary **PCIe x16 slot**. Check that all dedicated **PCIe 8-pin / 12VHPWR power cables** are fully locked in.
- **Step 4: Driver Clean Installation (Software Layer)**
  - Boot into Windows Safe Mode, run **Display Driver Uninstaller (DDU)** to completely wipe corrupted graphics drivers, restart, and perform a clean install of the latest official NVIDIA/AMD drivers.
- **Step 5: Stress Testing & Fix Confirmation**
  - Run a GPU stress benchmark (e.g., *FurMark* or *3DMark*) for 15–30 minutes while monitoring core temperatures via HWMonitor. If temperatures stay below 80°C and no artifacting or driver crashes occur, the fix is confirmed.

### Task 3: Storage Disconnect/Reconnect Simulation (SATA/NVMe)

- **Before Reconnecting (Drive Unplugged):**

- **POST / Boot Error Screen:** System halts with a black screen displaying:

*"Reboot and Select proper Boot device or Insert Boot Media in selected Boot device and press a key"*

or *"No bootable device found."*

- **BIOS Indicators:** Inside the UEFI/BIOS under the *Main* or *Boot Priority* menu, the M.2 NVMe or SATA port is listed as **None** or **Empty**, and the Windows Boot Manager option is missing.

- **After Reconnecting (Drive Reseated):**

- **BIOS Indicators:** The BIOS immediately detects the hardware drive string (e.g., Samsung SSD 980 1TB) with its correct capacity and restores Windows Boot Manager to Boot Option #1.
- **Boot Behavior:** The system bypasses the boot error prompt, loads the Windows spinning circle, and boots into the desktop OS normally.

#### Task 4: Random Restart Diagnosis (Component Selection & Steps)

- **Component to Check First: RAM (Memory)**
- **Reasoning:**
  - While GPUs cause driver crashes/artifacts and storage failures cause BSOD freezes or boot errors, **unstable or corrupted RAM directly corrupts running operating system kernel code**, causing sudden, unprompted reboots without warning. RAM is also the easiest and fastest component to test and isolate non-destructively.
- **Specific Diagnostic Steps for RAM:**
  1. **Visual & Physical Reseat:** Power off, clean the RAM contacts, and reseat each stick firmly into the recommended dual-channel slots (typically A2 and B2).
  2. **Single-Stick Isolation Test:** If two or more sticks are installed, remove all except one stick in Slot A2. Run the system under load. If it runs stably, test the second stick in the same slot to identify whether one specific stick is failing.
  3. **Disable XMP / EXPO Overclocking:** Enter BIOS and revert memory frequency from aggressive XMP/EXPO overclock profiles back to standard JEDEC defaults (e.g., 4800 MT/s or 2133 MHz) to check for memory controller instability.
  4. **Run Automated Memory Stress Diagnostics:** Create a bootable USB with **MemTest86** (or run the built-in *Windows Memory Diagnostic* by running mdsched.exe). Allow MemTest86 to complete at least 2 to 4 passes. If any red error addresses appear, the memory stick has hardware cell degradation and must be replaced.